Correlation and Pearson Correlation Coefficient

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1. Correlation Between Instagram Usage and Mobile Data Usage

Suppose the following data is collected from 8 friends:

Friend	Instagram Hours/Week (X)	Mobile Data Usage/Month (Y in GB)
1	5	8
2	7	10
3	10	13
4	12	15
5	15	18
6	18	21
7	20	23
8	22	25

Using a calculator or spreadsheet, the **Pearson correlation coefficient (r)** is approximately:

r ≈ 0.99

The value of **r** is very close to +1.

Interpretation:

There is a **strong positive correlation** between Instagram usage and mobile data usage. This means that, in this sample, friends who spend more time on Instagram generally use more mobile data.

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2. Pearson Correlation Formula

The Pearson correlation coefficient measures the **strength and direction of the relationship between two variables**.

Formula:

r=n∑XY−(∑X)(∑Y)[n∑X2−(∑X)2][n∑Y2−(∑Y)2]r = \frac{n\sum XY-(\sum X)(\sum Y)}{\sqrt{[n\sum X^2-(\sum X)^2][n\sum Y^2-(\sum Y)^2]}}

Where:

- **r** = Pearson correlation coefficient
- **X** = First variable
- **Y** = Second variable
- **n** = Number of observations

Interpretation of r:

Correlation Value	Interpretation
+1	Perfect positive correlation
0.7 to +1	Strong positive correlation
0.3 to 0.7	Moderate positive correlation
Around 0	Little or no correlation
-0.3 to -0.7	Moderate negative correlation
-0.7 to -1	Strong negative correlation
-1	Perfect negative correlation

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3. Correlation Between IMDb Ratings and BookMyShow Reviews

Suppose we collect data for 10 movies:

Movie	IMDb Rating (X)	BookMyShow Reviews (Y)
Movie 1	6.5	4,000
Movie 2	7.0	5,500
Movie 3	7.2	6,500
Movie 4	7.5	8,000
Movie 5	7.8	9,000
Movie 6	8.0	11,000
Movie 7	8.2	13,000
Movie 8	8.4	15,000
Movie 9	8.6	17,000
Movie 10	9.0	20,000

Using a spreadsheet or calculator, the Pearson correlation coefficient is approximately:

r ≈ 0.99

Interpretation:

The correlation is **strong positive**.

This means that in this sample, movies with higher IMDb ratings generally have a higher number of BookMyShow reviews. However, correlation does not necessarily mean that a higher IMDb rating directly causes more reviews.

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4. Zomato Orders and IPL Matches

Suppose we find a **strong positive correlation** between the number of food orders on Zomato and the number of IPL matches aired in a month.

This does **not necessarily mean that IPL matches directly cause more food orders**. Other factors such as weekends, holidays, major IPL matches, discounts, weather, and user behavior could also increase food orders.

Conclusion:

Correlation does not always mean causation. A relationship between two variables does not prove that one variable is directly causing the other.